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ACADEMIC VAULT

UNIZIK FRESHER EDITION | 2026

GST 104: History and Philosophy of Science - How Knowledge Actually Grows

From observation and falsification to African science and the ethics of discovery

GST / COMPULSORY

GST & Compulsory Courses

General Studies (GST / CED)
GST Directorate | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

Study smarter. Ask better questions. Pass with understanding.

Independent study support | Not an official UNIZIK publication

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Why this pack exists

This pack explains what makes science different from mere belief, how scientific ideas have changed across history, and how African knowledge systems contributed to medicine, metallurgy and agriculture. It also gives you the vocabulary for ethics-of-science questions.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Explain the scientific method, hypothesis, falsifiability, paradigm and theory without collapsing them into one word.
- Trace major shifts from ancient to modern science with accurate examples.
- Describe African contributions to science and technology using specific evidence.
- Discuss ethical issues such as research misconduct, informed consent and environmental responsibility.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

What science is - and is not

Science is a method of producing reliable knowledge through observation, measurement, hypothesis testing and public criticism. It is not a fixed body of facts, and it is not the same as technology.

A hypothesis is a testable prediction. A theory is a well-supported explanatory framework that has survived repeated testing. A law describes a regular relationship. Colloquial English uses 'theory' to mean a guess; science does not.

Falsifiability, associated with Karl Popper, means a claim must specify what evidence would count against it. 'Everything happens for a reason' is not scientific because no observation could refute it.

- Observation and measurement: record what happens under controlled conditions.
- Hypothesis: a specific, testable prediction.
- Experiment: manipulate variables while controlling others.

- Peer review: allow others to check methods, data and reasoning.
- Replication: repeat the study to see whether the result holds.

> **Correlation is not causation**
> Two things rising together may share a third cause, or one may cause the other in the opposite direction. Before claiming causation, look for controlled evidence and a plausible mechanism.

A short history of scientific change

Science has not progressed by simply accumulating facts. It has changed when old frameworks could no longer explain new observations.

- Ancient Greek natural philosophy: Aristotle's physics and the geocentric model.
- Islamic Golden Age: algebra, optics, hospitals and the preservation of Greek texts.
- Copernican revolution: the sun-centred model replaced geocentrism.
- Galileo and Newton: mathematised motion, telescopes and laws of gravitation.
- Darwin and Wallace: evolution by natural selection.
- Einstein: relativity transformed space, time and gravity.
- Watson, Crick, Franklin and Wilkins: the structure of DNA and modern molecular biology.

African science and technology

African societies produced empirical knowledge and technical innovation long before European contact. Recognising this is not sentiment; it is historical accuracy.

- Metallurgy: Nok iron smelting and Igbo-Ukwu bronze casting.
- Medicine: herbal pharmacopoeias, bone setting and knowledge of plant properties, some of which inform modern drug discovery.
- Agriculture: mixed cropping, soil management, terracing and indigenous seed selection.
- Mathematics: fractal-like patterns in African textiles and architecture; counting and weight systems.
- Astronomy: navigation and calendar systems based on celestial observation.

> **Avoid both extremes**
> Do not claim that all modern science was stolen from Africa, and do not claim that Africa had no science. Make specific, evidence-based claims about specific technologies.

Science, technology and society

Scientific knowledge is never automatically good. Its application depends on incentives, regulation and who bears the risk.

- Research ethics: informed consent, confidentiality, honest reporting and approval by an ethics committee.
- Misconduct: fabrication, falsification and plagiarism destroy trust and can harm patients.
- Dual use: the same research can produce medicine or weapons.